& CloudNova

Unified Cloud Infrastructure Management & Simulation Platform

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Services User Manual

Comprehensive Operational & Technical Guide to CloudNova Services

Compute & EC2 Engine	**S3 Object Storage**	**Multi-Engine DBs**
VPC Networks	**Live Telemetry**	**FinOps Billing**

SYSTEM INFORMATION & DEPLOYMENTS

- Live Web Application: https://cloud-nova-five.vercel.app/
- Backend API (Render): https://cloudnova-api.onrender.com
- Interactive Swagger Docs: https://cloudnova-api.onrender.com/docs
- Version & Security Status: v1.0.0 | Enterprise Scrypt & JWT Secured
- AWS Safe Mode Guard: ACTIVE (Zero Cloud Cost Simulation Mode)

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1. Platform Overview & 3-Tier Architecture

CloudNova is a production-ready, full-stack cloud management platform designed to unify resource provisioning, telemetry observability, and financial modeling across simulated and live cloud environments.

Architecture Stack

• **Presentation Layer:** React 19 with Vite 8. Highly optimized bundle served over Vercel Edge CDN with sub-second hot-reloads and responsive CSS layout.

• **Application API Layer:** FastAPI (Python 3.10+). Asynchronous ASGI architecture delivering sub-15ms response latency, strict Pydantic v2 data models, and automatic OpenAPI schema generation.

• **Data Storage Layer:** SQLite 3 with Write-Ahead Logging (WAL) and foreign key constraints, providing complete ACID compliance and multi-user relational safety.

• **Cloud SDK Gateway:** Amazon Web Services (AWS) Boto3 and Botocore Python SDK with automated development safeguards.

> **Security & Cost Zero-Risk Safeguard**
> CloudNova includes an integrated AWS safety lock (AWS_ENABLED=false) guaranteeing that all student and developer testing operates within a $0.00 cloud invoice ceiling.

2. Service 1: Authentication & Identity Management (IAM)

The CloudNova IAM service provides cryptographic authentication, user isolation, and brute-force intrusion prevention. Every cloud resource (VM, database, bucket, network) is bound strictly to the owner's identity.

Key Specifications & Technical Security

• **Password Hashing Algorithm:** Scrypt Key Derivation Function (KDF) using N=16384, r=8, p=1 and 16-byte cryptographically secure random salts. Stored as salt_hex:hash_hex.

• **Stateless Token Security:** JSON Web Tokens (JWT) using HMAC-SHA256 (HS256) encryption with a 30-minute expiration window.

• **Brute-Force Rate Limiter:** In-memory sliding window algorithm monitoring (IP, Email) pairs. Exceeding 5 failed attempts locks authentication for 15 minutes with HTTP 429 Too Many Requests.

Step-by-Step Usage Instructions

1. Open CloudNova at https://cloud-nova-five.vercel.app/. If not logged in, the Authentication screen opens.
2. Click "Sign Up" at the bottom of the card.
3. Enter your Full Name, Email Address, and Password (minimum 8 characters).
4. Click "Sign Up" to register. The backend verifies uniqueness and saves the encrypted credentials.
5. Log in with your email and password. Upon successful verification, your JWT token is stored in localStorage.

REST API Endpoints Reference

```
POST /signup                - Body: { name, email, password } -> Creates account
POST /login                 - Body: { email, password } -> Returns { access_token,
token_type }
GET  /me                    - Headers: Authorization: Bearer <token> -> Returns user
identity
```

3. Service 2: Compute & Virtual Machines (EC2 Engine)

The Compute service provides complete lifecycle control over virtual machine instances and generates validated AWS EC2 instance deployment previews.

Supported Virtual Machine Specifications

• **t3.micro:** 2 vCPUs, 1.0 GiB RAM, baseline burst performance. Ideal for microservices and test web servers.

• **t3.small:** 2 vCPUs, 2.0 GiB RAM, network bursts up to 5 Gbps. Suitable for staging and moderate databases.

• **t3.medium:** 2 vCPUs, 4.0 GiB RAM, dual vCPU capacity. Recommended for production application workloads.

AWS EC2 Instance Preview Parameters

• **Instance Name:** Custom identifier adhering to AWS naming guidelines (e.g. Web-Cluster-Prod-01).

• **Geographic Regions:** ap-south-1 (Mumbai), ap-south-2 (Hyderabad), us-east-1 (N. Virginia).

• **Operating System:** Ubuntu Server 24.04 LTS (x86_64 architecture).

• **Storage Disk:** General Purpose SSD (gp3) volume sizing from 8 GB up to 100 GB.

Step-by-Step Usage Guide

1. Navigate to "Compute" (Ø=Ý¥ Compute) in the sidebar or mobile menu.
2. Review the summary metrics: Total Machines, Running instances, and Stopped instances.
3. To stop a running machine: Click the blue "Stop" button. The status switches to "%ï Stopped".
4. To restart a machine: Click "Start". The status immediately updates to "%ï Running".
5. To delete a machine: Click "Delete" and confirm the deletion dialog.
6. To configure a new EC2 instance: Click "+ Configure EC2". Enter the instance name, choose the region and instance type, enter disk size in GB, and click "Generate EC2 Preview".
7. The system generates an immediate validation report with estimated hourly and monthly operating costs.

REST API Endpoints Reference

```
GET    /vms                  - Returns list of user virtual machines
POST   /vms                  - Body: { name, size } -> Creates new VM
PATCH  /vms/{id}             - Body: { status } -> Updates VM status (Running/Stopped)
DELETE /vms/{id}             - Permanently deletes VM record
POST   /aws/ec2/preview      - Body: { name, region, operating_system, instance_type,
volume_gb }
```

4. Service 3: Cloud Object Storage (Buckets Service)

The Cloud Storage service mimics Amazon S3 and Google Cloud Storage buckets, enabling organized cloud storage quotas with dynamic capacity calculation.

Storage Analytics Engine

- **Total Buckets:** Dynamic tally of all active storage buckets assigned to the authenticated user.
- **Total Storage Capacity:** Cumulative sum of all allocated storage gigabytes across the fleet.
- **Average Bucket Size:** Calculated arithmetic mean (Total Storage GB ÷ Total Buckets).

Step-by-Step Usage Guide

1. Click "Storage" (%¤ Storage) from the navigation panel.
2. Inspect the analytics cards at the top of the view.
3. In the "Create Bucket" form, enter a unique bucket name (e.g., "customer-invoices-2026").
4. Enter the storage size in gigabytes (e.g., 50).
5. Click "Create Bucket". The backend persists the bucket and instantly refreshes total capacity.
6. To remove an obsolete bucket, click "Delete" on the target bucket row.

REST API Endpoints Reference

```
GET    /storage/buckets      - List all buckets with allocated size_gb
POST   /storage/buckets      - Body: { name, size_gb } -> Allocates new bucket
DELETE /storage/buckets/{id} - Removes bucket from database
```

5. Service 4: Multi-Engine Database Provisioning

CloudNova provides multi-engine database cluster management supporting relational SQL and unstructured NoSQL document architectures.

Supported Database Engines

- **PostgreSQL:** Enterprise object-relational SQL database with strong consistency and JSONB support.
- **MySQL:** High-throughput relational SQL database optimized for web workloads and read scalability.
- **MongoDB:** Distributed document-oriented NoSQL database supporting flexible BSON/JSON schemas.
- **SQLite:** Lightweight serverless embedded relational engine for micro-services and testing.

Operational Lifecycle Management

1. Open "Databases" (%£ Databases).
2. Enter a Database Name (e.g. "inventory_db") and select an engine.
3. Click "Create Database". The new database appears in the overview table.
4. Toggle database availability: Click "Deactivate" to suspend connections, or "Activate" to restore.
5. Click "Delete" to permanently drop the database cluster.

6. Service 5: VPC Virtual Networking & IP Routing

The Networking service provisions software-defined Virtual Private Clouds (VPC) with strict IPv4 CIDR subnet validation and multi-region routing.

Pre-Validated Network CIDR Ranges

• **10.0.0.0/16:** Class A private IP block (65,536 addresses). Default for corporate enterprise cloud architectures.

• **172.16.0.0/16:** Class B private IP block (65,536 addresses). Suitable for staging clusters and development suites.

• **192.168.0.0/24:** Class C private subnet (256 addresses). Optimal for microservices and isolated testing environments.

Subnet Validation Engine

All submitted CIDR notations are parsed through Python's native ipaddress.ip_network() library. Invalid IP masks or malformed addresses are automatically rejected before database insertion.

Step-by-Step Usage Guide

1. Navigate to "Networking" (Ø<ß Networking).
2. Enter your Network Name (e.g., "prod-vpc-mumbai").
3. Select your CIDR range from the dropdown menu.
4. Choose your deployment region (Asia Pacific, US East, or Europe).
5. Click "+ Create Network". The validated VPC is stored and active.
6. Toggle between "Active" and "Inactive" status, or delete expired networks.

7. Service 6: Infrastructure Monitoring & Telemetry

Provides real-time system observability. Telemetry values update dynamically every 3 seconds to reflect fluctuating computational pressure.

• **CPU Usage (%):** Real-time microprocessor consumption with dynamic load simulation (20% – 70%).

• **Memory Usage (%):** Active RAM utilization tracker with capacity alerts (50% – 80%).

• **Network Traffic (Mbps):** Inbound and outbound bandwidth throughput monitoring (10 – 80 Mbps).

• **Server Uptime (%):** Service availability tracker maintained at 99.9% uptime SLA.

8. Service 7: FinOps, Billing & Cost Estimation

Enables cloud financial management (FinOps) by attributing estimated monthly costs in Indian Rupees (¹).

Service Category	Monthly Cost (¹)	Percentage
Compute Engine	¹750	60.0%

Cloud Storage	¹300	24.0%
Managed Databases	¹200	16.0%
TOTAL ESTIMATED SPEND	¹1,250 / month	100.0%

9. Service 8: AWS EC2 Simulation & Cloud Gateway

CloudNova integrates with Amazon Web Services through the official Boto3 SDK, guarded by an automatic simulation switch.

Safe Mode vs. Live AWS Mode

• **Simulation Mode (AWS_ENABLED=false):** Default operational mode. Evaluates instance configurations, valid operating systems, and disk volumes without invoking live AWS billing APIs. Guaranteed $0 cloud invoice.

• **Live Connected Mode (AWS_ENABLED=true):** Uses configured AWS IAM credentials to query active AWS region endpoints via ec2.describe_regions().

10. Cross-Device Responsiveness (Mobile & Laptop)

CloudNova is engineered using mobile-first CSS architecture with fluid viewport typography:

• **Desktop & Laptop (Width >= 992px):** Permanent docked navigation sidebar (250px), multi-column resource overview cards, spacious tables, and instant menu access.

• **Smartphone & Tablet (Width < 992px):** Full-width 100% viewport utilization. Sidebar transitions into an off-canvas slide-out drawer accessible via the top hamburger button (&0). Includes dark backdrop blur, touch-close button ('), and horizontal scrollable tables (-webkit-overflow-scrolling: touch).

11. Troubleshooting & Frequently Asked Questions (FAQ)

• **Q: Topbar shows "API: Offline"?:** Render free-tier instances sleep when inactive. Please allow 30–45 seconds for backend cold start, then refresh the browser.

• **Q: How do I test on my smartphone?:** Open https://cloud-nova-five.vercel.app/ on your phone browser. Tap the &0 icon in the top header to reveal the navigation menu.

• **Q: Will generating EC2 instances charge my card?:** No. CloudNova operates in zero-cost simulation mode by default. No real instances are created on AWS.

• **Q: How do I run automated unit tests?:** Run "pytest backend/tests -v" from the repository. All 23 automated unit and integration tests will execute.